

Pre-Board Exam(2024-25)

Class-X

Time-2 Hours

Subject :Artificial Intelligence(417)

MM-50

General Instructions:

1. Please read the instructions carefully.
2. This Question Paper consists of 21 questions in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. **Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.**
5. All questions of a particular section must be attempted in the correct order.
6. **SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**
 - i. This section has 05 questions.
 - ii. Marks allotted are mentioned against each question/part.
 - iii. There is no negative marking.
 - iv. Do as per the instructions given.
7. **SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**
 - i. This section has 16 questions.
 - ii. A candidate has to do 10 questions.
 - iii. Do as per the instructions given.
 - iv. Marks allotted are mentioned against each question/part.

SECTION A: OBJECTIVE TYPE QUESTIONS

Q. NO.	QUESTION		MARKS
Q. 1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)		
i.	(a)		1
ii.	(c)		1
iii.	(c)		1
iv.	(c)		1
v.	(b)		1
vi.	(d)		1

Q. 2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)		
i.	(a) Both A and R are true, and R is the correct explanation of A.		1

ii.	(b)		1
iii.	(c)		1
iv.	(b)		1
v.	(b)		1
vi.	c)		1

Q. 3	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)		
i.	False		1
ii.	(a)		1
iii.	(b)		1
iv.	(a)		1
v.	Script bot		1
vi.	(a)		1

Q. 4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)		
i.	(c)		1
ii.	Text Classification		1
iii.	(b)		1
iv.	(c)		1
v.	(b)		1
vi.	Term Frequency Inverse Document Frequency		1

Q. 5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)		
i.	(c)		1
ii.	(b) Precision and Recall		1
iii.	(b)		1
iv.	(a)		1
v.	True		1
vi.	(c) False Negative		1

SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks)

Answer each question in 20 – 30 words.

Q. 6	A business plan is a written document that outlines a company's goals, strategies, and financial projections. It's a roadmap for a business's growth and development, and is important for both startups and existing businesses.	2
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	(1 mark for each point)	
Q. 7	Stress management is important because it can help you maintain your physical and mental health, and improve your overall well-being: Avoid health problems: Chronic stress can lead to serious health problems, including depression, anxiety, diabetes, and heart disease. Improve your mood: Stress management can help you feel better and get along better with others. Improve your sleep: Stress management can help you sleep better. (1 mark for each point)	2
Q. 8	Two methods to protect our data on the computer: 1. Use passwords to login to your computer. 2. Install Anti-virus and Firewall 3. Encrypt Data 4. Secure sites (1 mark for each correct method, 1*2=2)	2
Q. 9	I. Fulfill Customer Needs II. Use Local Materials III. Help Society IV. Create Jobs V. Sharing of Wealth VI. Lower Price of Products (1 mark for each correct point, 1*2=2)	2
Q. 10	This stage is the Survive stage of an entrepreneur's career process. In this stage, even if there are many entrepreneurs in the market, the new entrepreneur has to remain in a competitive market. (1 mark for mentioning the stage; 1 mark for correct explanation)	2

Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)

Q. 11	Face Lock, Google Maps, Google Assistant, E-Commerce apps, Social Media apps, Audio/Video Streaming Platforms and etc.	2
Q. 12	Rule Based Approach: It refers to the AI modelling where the relationship or patterns in data are defined by the developer. The machine follows the rules or instructions mentioned by the developer, and performs its task accordingly. For example, suppose you have a dataset comprising of 100 images of apples and 100 images of bananas. To train your machine, you feed this data into the machine and label each image as either apple or banana. Now if you test the machine with the image of an apple, it will compare the image with the trained data and according to the labels of trained images, it will identify the test image as an apple. This is known as Rule based approach. The rules given to the machine in this example are the labels given to the machine for each image in the training dataset.	2
Q. 13	Steps of AI project life cycle: 1. Problem Scoping 2. Data Acquisition 3. Data Exploration 4. Modelling 5. Evaluation (½ mark for mentioning each stage, ½ *4=2)	2

Q. 14	Resolution refers to the number of pixels in an image, usually measured in pixels per inch (PPI). Higher resolution means more pixels and greater detail, resulting in better image quality.	2													
Q. 15	Stemming Lemmatization happily happi happy Process that takes longer time for execution is lemmatization (½ marks each for identifying the correct stem and lemma; 1 mark for identifying the correct process)	2													
Q. 16	<table><tr><th colspan="2" rowspan="2">Confusion matrix</th><th colspan="2">Reality</th></tr><tr><th>YES</th><th>NO</th></tr><tr><th rowspan="2">prediction</th><th>YES</th><td>150</td><td>40</td></tr><tr><th>NO</th><td>50</td><td>60</td></tr></table>	Confusion matrix		Reality		YES	NO	prediction	YES	150	40	NO	50	60	2
Confusion matrix				Reality											
		YES	NO												
prediction	YES	150	40												
	NO	50	60												

Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)

Q. 17	Artificial Intelligence (AI) refers to any technique that enables computers to mimic human intelligence i.e., make decisions, predict the future, learn and improve on its own. With respect to the type of data fed in the AI model, AI models can be broadly categorised into three domains: 1. Data sciences 2. Computer vision 3. Natural Language Processing Data Science takes input in the form of numeric and alphanumeric data. Computer Vision takes input in the form of images and videos. Natural Language Processing takes input in the form of text and speech. (1 mark for definition of AI; ½ mark each for the names of the domains; ½ mark each for the type of data input to domains)	4
Q. 18	(A) Project Cycle is a step-by-step process to solve problems using proven scientific methods and drawing inferences about them. The AI Project Cycle provides us with an appropriate framework which can lead us towards the goal. The AI Project Cycle mainly has 5 stages: They are a) Problem Scoping b) Data Acquisition c) Data Exploration d) Modelling e) Evaluation. (B)a. Who, b. What c. Where d. Why	4
Q. 19	a) Supervised learning is an approach to creating artificial intelligence (AI), where the program is given labelled input data and the expected output results. OR Supervised learning is a learning in which we teach or train the machine using data which is well labelled that means some data is already tagged with the correct answer. After that, the machine is provided with a new set of examples (data) so that supervised learning algorithm analyses the training data (set of training examples) and produces a correct outcome from labelled data. In a supervised learning model, the dataset which is fed to the machine is labelled. It means some data is already tagged with the correct answer. In	4

	other words, we can say that the dataset is known to the person who is training the machine only then he/she is able to label the data. b) Unsupervised Learning: An unsupervised learning model works on unlabelled dataset. This means that the data which is fed to the machine is random and there is a possibility that the person who is training the model does not have any information regarding it. The unsupervised learning models are used to identify relationships, patterns and trends out of the data which is fed into it. It helps the user in understanding what the data is about and what are the major features identified by the machine in it.		
Q. 20	1. Tokenisation Akash, and, Ajay, are, best, friends Akash, likes, to, play, football, but, Ajay, prefers, to, play, online, games 2. Removal of stopwords Akash, Ajay, best, friends Akash, likes, play, football, Ajay, prefers, play, online, games 3. converting text to a common case akash, ajay, best, friends akash, likes, play, football, ajay, prefers, play, online, games 4. Stemming/Lemmatisation akash, ajay, best, friend akash, like, play, football, ajay, prefer, play, online, game (1 mark for each step; 1*4=4)		4
Q. 21	(i) TP=60, TN=10, FP=25, FN=5 60+25+5+10=100 total cases have been performed (ii) (Note: For calculating Precision, Recall and F1 score, we need not multiply the formula by 100 as all these parameters need to range between 0 to 1) Precision = $TP/(TP+FP)$ $=60/(60+25)$ $=60/85$ $=0.7$ Recall = $TP/(TP+FN)$ $=60/(60+5)$ $=60/65$ $=0.92$ F1 Score = $2*Precision*Recall/(Precision+Recall)$ $=2*0.7*0.92/(0.7+0.92)$ $=0.79$ (1 mark for total number of cases; 1 mark each for the calculation of precision, recall and F1 score)		4